

The Empathetic Fitting Room: A Physiologically Adaptive Retail Solution

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Abstract—Fitting rooms are a crucial conversion point in fashion retail, yet 36% of consumers report leaving a shop without purchasing due to negative in-store experiences. This paper proposes a physiologically adaptive fitting room prototype that adjusts lighting and audio in real time based on live heart rate (via RR-intervals), aiming to reduce user stress. The system connects a Polar H10 sensor to Philips Hue lights and a four-stem audio engine via an asynchronous Python pipeline, using a PID controller to produce a continuous reaction score driving both outputs. A within-subjects evaluation with eleven participants compared the adaptive condition against a static control. The system demonstrated strong proportional responsiveness to physiological state ($r=0.98$) and maintained real-time performance, with 99.5% of processing cycles completing within the 400 ms threshold required for biofeedback. Bayesian paired t-tests showed very strong evidence of greater subjective stress reduction under the adaptive condition ($BF_{10}=53.882$); a physiological (RR-interval) effect was not detected at the group level ($BF_{10}= 0.302$), presumably due to the brief evaluation window. Despite the lack of group-level physiological effects, the significant subjective benefits and successful technical implementation establish a proof of concept for heart rate-driven adaptive retail environments.

Keywords—sensory marketing, experiential retail, heart rate, ambient intelligence, biofeedback, PID control, Internet of Things (IoT), affective computing

I. INTRODUCTION

As Gen Z and Alpha are expected to represent 40% of fashion buyers by 2035, fashion companies need to prioritise retail experiences over loyalty (Petro, 2026; Barton *et al.*, 2025). Brands such as Rebecca Minkoff have already explored the opportunities of interactive fitting rooms, allowing shoppers to manually control lighting, request different sizes via RFID-tagged items, and browse styling suggestions, merging customers' online and offline shopping experiences to drive sales (Hendriksz, 2014; Stromberg, 2018).

Fashion shopping behaviour is highly driven by emotion (Amed *et al.*, 2025). Lighting and music are key elements in transforming retail spaces into sensory destinations, triggering emotional responses from customers and encouraging them to stay, interact and come back (Shtompel, 2025; Katz, 2024). This becomes crucial in fitting rooms: in-store trial reduces return rates to under one in ten, compared to nearly one in five for online purchases (AlertTech, 2016), proving the fitting room's role as a critical point in purchase decisions. Despite this, 39% of consumers across Europe report negative feelings when using fitting rooms, and 36% abandon a purchase due to negative in-store experiences, rising to 41% in the UK (Zalando, 2024).

Existing “smart” fitting rooms such as Rebecca Minkoff's rely entirely on manual inputs rather than responding to a customer's emotional state. Affective computing offers a more seamless alternative, as it can recognise feelings via heart rate and breathing patterns (Timashev, 2025), however this opportunity has not yet been explored in the context of fitting rooms. To address this gap, the present project aims to design, build, and evaluate a bio-responsive fitting room prototype that automatically adjusts lighting and audio based on a user's real-time heart rate to reduce perceived stress. To achieve this aim, the project will fulfill the following objectives:

- 1) Review literature on sensory marketing, HRV-based stress measurement, and bio-responsive IoT architecture;
- 2) Conduct qualitative research to establish consumer design requirements;
- 3) Design and implement a closed-loop, sub-400ms PID control architecture connecting live physiological data to synchronized multimodal (lighting and audio) outputs;
- 4) Empirically evaluate both the subjective efficacy and the proportional responsiveness of the control architecture using Bayesian statistical analysis.

II. LITERATURE REVIEW

A. Experiential Retail & Sensory Marketing

Experiential retail prioritizes a highly enjoyable consumer experience, allowing shoppers to form an emotional connection with a brand (Servais *et al.*, 2022). Consequently, brands utilise sensory marketing to influence how customers feel about a certain product and what they choose to buy by stimulating their senses, such as sight and hearing (Krishna, 2012). The psychological mechanism behind this influence can be explained through the framework of consumer inference, according to which consumers rely on available environmental cues to form broader conclusions about a brand's value (Kardes *et al.*, 2004).

While carefully designed multi-modal sensory feedback (auditory and visual) can reduce cognitive load and restore future purchase intention (Reynolds-McIlroy and Morrin, 2019), highly stimulating environments often trigger sensory overload (Doucé and Adams, 2020). It can therefore be argued that a significant use of high-arousal stimuli in retail spaces actively harms the consumer's cognitive state once they enter the confined, evaluative space of a fitting room. This may help explain the high rates of fitting room frustration identified in industry research (Zalando, 2024),

where sensory overload in an evaluative context may directly contribute to abandoned purchases.

B. Affective Computing & HRV

Picard (1997) defined “affective computing” as “computing that relates to, arises from, or deliberately influences emotions” (p. 3). Within this field, Heart Rate Variability (HRV) – a standard measure of cardiac vagal activity, quantified through the analysis of beat-to-beat (RR) intervals (Porges, 2007) – is a powerful real-time indicator of a user’s emotional state (Motogna *et al.*, 2021). Because acute stress activates the sympathetic “fight-or-flight” reaction, heart rate increases and HRV decreases (Jerath *et al.*, 2023), making low HRV associated with anxiety (Goessl *et al.*, 2017). However, HRV measurement is not without limitations: motion artefacts can corrupt RR interval readings, artificially inflating or suppressing HRV values and producing unreliable stress estimates (Georgiou *et al.*, 2018). This limitation is especially relevant to retail: unlike controlled settings where most HRV research occurs, fitting rooms are active environments where natural movement makes data collection more challenging. This constraint is methodologically significant for the present study, as it influences the decision to ask participants to remain stationary during prototype evaluation.

C. Bio-Responsive Environments

The preceding sections establish two converging problems: sensory environments can either support or harm the consumer, and HRV offers a reliable but underused real-time signal of which state a customer is in; Aml (Ambient Intelligence), which Alhamid *et al.* (2012) define as electronic spaces that detect and react to human presence, addresses this convergence directly. Building on this concept, Leonidis *et al.* (2019) built CaLmi, an Aml living room that utilizes wearable electrodermal activity (EDA) and blood volume pressure (BVP) sensors to trigger synchronized multisensory audio-visual interventions; in a small-sample evaluation, participants’ EDA values decreased by an average of 49% following multisensory sessions, and the majority of participants were calmer after these sessions compared to visual-only interventions. It can however be argued that, in the context of this project, EDA/BVP sensors are highly susceptible to motion artifacts, making them impractical for an active fitting room environment.

Furthermore, while traditional Aml environments mostly rely on pre-recorded media, recent affective computing studies such as those by Idrobo-Ávila *et al.* (2022) have mapped HRV directly into generative musical sequences, representing a new opportunity of physiologically responsive audio design.

A key design consideration for bio-responsive systems is the control architecture that translates physiological signals into environmental outputs. Fairclough (2009) presents physiological computing as a continuous biocybernetic loop, where the system adapts to the user’s state over time. This is essential for stress recovery specifically: McEwen (1998) shows that stress-related physical damage results from chronic physiological arousal (allostatic load) rather than isolated events.

Overall, this progression from static interventions to fully generative HRV-to-music mappings has occurred exclusively in domestic or experimental contexts; no study has applied real-time physiologically adaptive multimodal biofeedback to a retail fitting room, representing the gap this project addresses.

D. IoT Latency & Architecture

For a bio-responsive environment to feel reactive, the system must process each heartbeat and update the lights and audio quickly enough that the change feels connected to the user’s physiological state. Research suggests this requires end-to-end system latency to remain below 400 milliseconds (ms) (Ng *et al.*, 2020). Achieving this in practice is challenging because Bluetooth Low Energy (BLE) signals in enclosed indoor spaces are susceptible to interference and packet loss, introducing unpredictable delays (Ng *et al.*, 2020).

In a traditional synchronous program, reading from the Bluetooth sensor and sending commands to the lights and speakers would occur sequentially, with each operation blocking the next and causing the environment to fall out of sync with the user’s live heartbeat. The software therefore must handle the sensor stream and environmental commands concurrently, motivating the use of Python’s asynchronous programming library (*asyncio*). By using *asyncio*’s single-threaded event loop rather than multi-threading, the system avoids the overhead of thread synchronization (Morales *et al.*, 2025), making it an efficient choice for prototype development.

E. Research Questions and Hypotheses

Addressing the identified gaps in the literature review, this study raises the below research questions:

- *RQ1*: Does a bio-responsive environment adapting lighting and audio based on heart rate reduce perceived user stress compared to a static environment?
- *RQ2*: Does a continuous adaptive control architecture produce a proportional environmental response that reflects the user’s physiological stress trajectory throughout the session?

Based on these research questions, the following hypotheses are proposed:

- *H1*: A bio-responsive environment that adapts lighting and audio in real time based on heart rate will produce a greater reduction in perceived stress than a static control environment.
- *H2*: A PID control architecture will produce a continuous, proportional environmental response rather than a binary threshold shift.

III. USER RESEARCH

To identify the design requirements for a fitting room that prevents user frustration, a primary qualitative study was conducted. Purposive sampling was used to recruit 7 participants, with a skew toward Gen Z to reflect future retail demographics. While Gen Z encompasses individuals born between 1997 and 2012 (Dimock, 2019), ethical considerations regarding minors restricted the sample to participants aged 18 and older (born 1997-2007). Table I shows an overview of the sample. Ethical approval was secured prior to data collection, ensuring informed consent and strict data anonymization. The participant information sheet for this phase, together with the consent form template, is under Appendix A.

TABLE I. USER RESEARCH SAMPLE

Category	Details
Genders	4 Females, 3 Males
Generation	6 Gen Z's over 18 (born 1997-2007), 1 Millennial (born 1981-1996; Dimock, 2019)
Occupations	1 Employed Full-Time, 6 Students

Questions for the semi-structured interviews were developed starting from the research questions (see Appendix B). The thematic analysis revealed a mismatch between consumer needs and current retail offerings. Key themes included:

- The Fitting Room as a Chore: Participants viewed fitting rooms as stressful, citing rushing, poor hygiene, and flimsy curtains.
- Sensory Overload: The loud, fast-paced music of fast-fashion stores caused overstimulation and anxiety. Participants desired calming interventions like ambience music, "white noise" or "jazz".
- Psychological Impact of Lighting: Unflattering, top-down lighting was identified as a primary trigger for body image insecurities and abandoned purchases.
- The Wearable Tech Barrier: While participants found chest-strap sensors physically impractical for changing clothes, they all accepted anonymous biometric tracking (Data Privacy), provided there was a premium sensory payoff.

The full thematic analysis, including themes, codes, and further descriptions, is included in Appendix C.

IV. SYSTEM DESIGN

A. Design Requirements

The findings from the thematic analysis informed three main design requirements for the prototype:

1. Adaptive, layered ambient sound that simplifies as stress increases;
2. Warmer, softer lighting during elevated stress, brighter and cooler as the user calms;
3. A biofeedback loop responsive enough to justify wearing a chest strap sensor.

B. Hardware Selection

The prototype uses three pieces of hardware managed by a single laptop, which runs the asynchronous Python pipeline described in the next section and sends signals to both lights and speakers simultaneously. The sensor and light connect via BLE, while audio is delivered through wired speakers.

- The Polar H10 chest strap was selected as the heart rate sensor for its high-fidelity RR-interval data via the *bleakheart* driver (Smeraldi, 2025).
- A Philips Hue light strip was chosen as the lighting output, controlled directly over BLE via the *HueBLE* library, removing the need for a smart home bridge (HueBLE, no date). The light strip was positioned behind a mirror rather than overhead, addressing the user research finding that top-down illumination was a trigger for body image anxiety in fitting rooms. Fig. 1 shows the planned set-up.

- Audio output is delivered through standard wired speakers connected to the laptop, which runs the *pygame.mixer* engine.



Fig. 1. Planned prototype fitting room set-up.

C. Software Architecture

To ensure the environment stays synchronized with the user's live heartbeat, the core software was built using Python's *asyncio*. As the Bluetooth sensor detects heartbeats, it places them into an *asyncio.Queue*, which acts as a buffer to ensure no data is lost while the system is temporarily busy updating the lights and music. The software continuously reads from this queue, averaging the last 8 heartbeats to filter out sensor noise. Before the system becomes active, it collects a 20-beat calibration sample to learn the user's normal resting baseline, ensuring the experience is tailored to each individual's physiology (unless exceeding 95 bpm, triggering a 75 bpm fallback).

During live operation, the system continuously compares the user's current heart rate against their personal baseline. The difference is processed by a Proportional-Integral-Derivative (PID) controller. A PID controller is a feedback control mechanism traditionally used to regulate a system toward a desired target value by continuously measuring the error $e(t)$ between the setpoint and the actual value (Jiang, 2024). The controller output $u(t)$ is calculated continuously over time t using the following formula (Cheung, 2020):

$$u(t) = K_p e(t) + K_i \int e(\tau) d\tau + K_d \dot{e}(t)$$

This represents the weighted sum of three terms, which produces a stabilised score from -100 (highly stressed) to +100 (highly calm). This reaction score $u(t)$ is then mapped continuously to both the lighting and audio outputs. The three terms are:

- The Proportional component generates an output scaled directly to the current error (Jiang, 2024). Here, the error represents the deviation of the user's current RR-interval from their personal baseline, meaning the Proportional responds proportionately to the user's current stress (scaled to a 150 ms maximum sensitivity threshold). A high proportional weight ($K_p=0.8$) allows the system to react immediately to acute stress.
- The Integral term accumulates error over time, responding to sustained deviation from the setpoint (Jiang, 2024). With $K_i=0.05$, the accumulated error is capped at ± 50 , limiting the Integral term's contribution to just ± 2.5 points at any one moment.



This contribution is small, but it builds up and persists for as long as the stress continues, unlike the Proportional term, which resets at every new heartbeat. This gives the system memory of sustained stress, while the cap keeps the effect subtle and prevents excessive build-up, so the system stays responsive when the user relaxes.

- The Derivative term reacts to how quickly the error is changing, attenuating the controller output when the error is fluctuating rapidly (Jiang, 2024). The 8-beat rolling average described above already reduces raw beat-to-beat noise before it reaches the controller; the Derivative term ($K_d=0.3$) provides a second layer of damping on top of this, based on the rate of error change, to suppress any residual single-beat artefacts that survive the averaging.

Moreover, a real-time monitoring interface (Fig. 2) was implemented using Python's *Tkinter* library to allow the researcher to observe live physiology, PID output, and environmental values during evaluation sessions. An initial version was generated with AI assistance and then used as a reference during development of the final interface (see Appendix F for the Generative AI statement). To prevent conflicts between *Tkinter* and *asyncio*, the graphical interface operates as an independent process. It retrieves data by reading a shared JSON file every 500 ms, ensuring the main sensor and actuator pipeline remain uninterrupted.

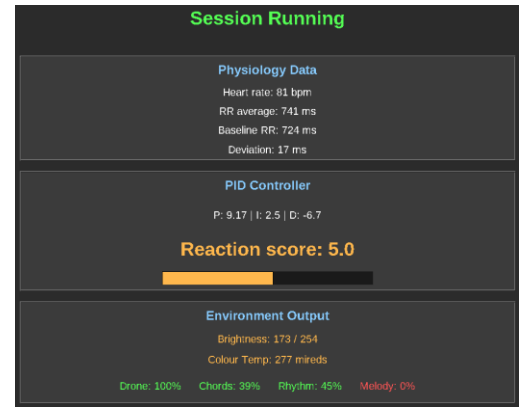


Fig. 2. Live monitor interface.

A secondary breathing modulation layer was implemented to subtly sync the room to the user's natural breathing. Because a person's heart rate speeds up slightly when they inhale and slows down when they exhale (Yasuma and Hayano, 2004), the system can track this respiratory rhythm. It does this by taking a very fast average of the user's last four heartbeats and comparing it against the slower eight-beat average used by the main system. This breathing effect only changes brightness (colour temperature and music remain controlled solely by the PID output) and was observed informally during sessions but not formally measured; this is discussed further under Future Work. The full pipeline is illustrated in Fig. 3.

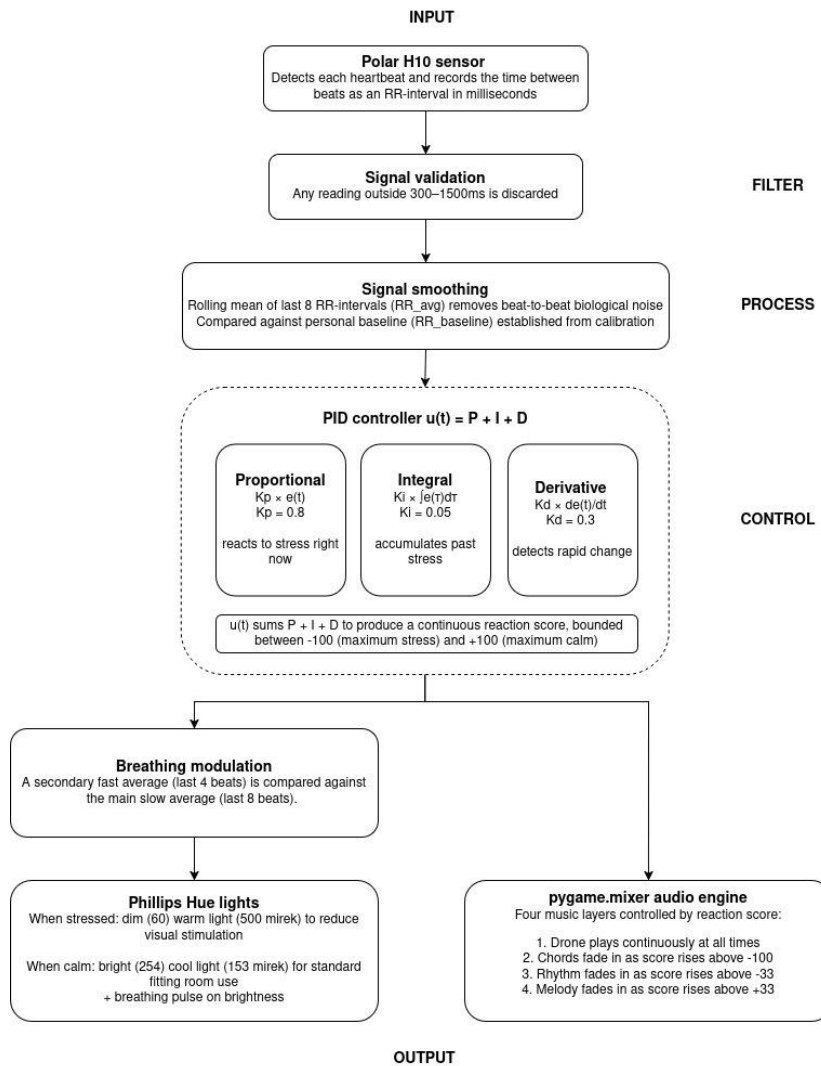


Fig. 3. System block diagram illustrating the closed-loop bio-responsive pipeline.

Several control strategies were considered before selecting the PID approach: Fig. 4 shows the different

approaches considered, based on the researcher's own testing data.

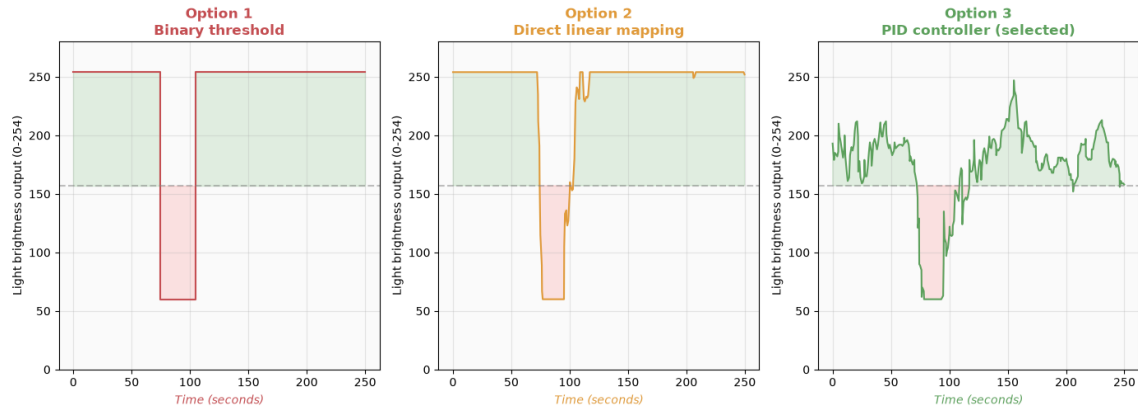


Fig. 4. Comparison of control strategies.

The simplest alternative was a binary threshold design, in which the system switches between a "stressed" and a "calm" state when the RR-interval crosses a fixed value, however this produces abrupt switching at the threshold crossing. The green and red shaded areas represent periods where the resulting light output falls above or below the midpoint of the brightness range, respectively. Applying a rolling average to the input before the threshold would reduce the flickering caused by natural beat-to-beat heart rate variation. A participant who crossed the threshold a single heartbeat ago, though, would receive the same environmental response as one who had been stressed for five minutes, which is physiologically inappropriate for a corrective biofeedback intervention. Effective physiological computing relies on a continuous 'biocycbernetic loop' where the system dynamically adapts to the user's ongoing state (Fairclough, 2009). An initial prototype using a binary threshold design was implemented and rejected following observed flickering during testing.

A direct linear mapping, where the RR-interval is proportionally scaled to an output value with no temporal memory, improves upon this; it removes the flickering but responds identically to brief and sustained deviations. Since it contains no integral term and therefore retains no memory of prior states, each input is processed independently of what came before. This means the system treats a momentary spike in stress the same as prolonged anxiety. This is a significant flaw for stress recovery, as distinguishing between brief and sustained stress is clinically crucial (McEwen, 1998).

The PID controller overcomes these limitations by combining graded response, memory of sustained stress, and damping of rapid fluctuations into a smooth, continuous output (as described above). Because simpler methods cannot provide smooth adjustments, long-term memory, and stability all at once, the extra complexity of the PID approach is justified. Note that although the PID transitions appear abrupt in the figure due to the compressed 250-second time axis, the physical system translates these signals into gradual crossfades, ensuring the environmental changes remain subtle to the user. Moreover, although the figure illustrates a comparison using the lighting output, the same control-strategy characteristics apply equally to the audio-mixing layer, which is driven by the same PID score.

D. Biofeedback UX Design

The system employs a corrective biofeedback philosophy, altering the environment to encourage physiological

recovery. To translate the PID's reaction score into physical outputs, the score is first converted into a percentage between 0% (maximum stress) and 100% (maximum calmness). Lighting is driven directly by this percentage, mapped proportionally to base brightness (60-254, further modulated by a ± 40 breathing variance) and colour temperature (500-153 mireds): as the calm percentage rises, brightness increases and colour temperature decreases, producing dim, warm light under stress and bright, cool light, consistent with standard fitting room conditions as the user calms. Audio, by contrast, is driven directly by the raw reaction score rather than the percentage: a four-stem dynamic mixing engine built with *pygame.mixer* plays a drone, soft chords, a rhythm, and a melody in perfect sync. The drone (Danke, 2023) plays continuously as a constant base throughout the session; as the score rises from stressed (-100) toward calm (+100), the system functions as an automated mixing desk, fading in chords (Fxsks, 2025) above -100, rhythm (Dakotalocals, 2018) above -33, and melody (Prodbkg1, 2022) above +33, so the soundscape becomes progressively richer as the user calms.

While the PID controller's gains were stabilised during the researcher's initial self-testing, its target output could still shift substantially when encountering the unpredictable physiological variance of different evaluation participants. To prevent such transitions from reaching the light output, both brightness and colour temperature are updated incrementally and driven directly by the PID output: with every heartbeat, the system moves the current lighting state one small step toward the PID controller's current target, rather than shifting to it directly. This ensures the room transitions gradually over several seconds instead of changing instantly to a new state.

V. SYSTEM EVALUATION

A. Experimental Design & Conditions

A within-subjects design was used, meaning all participants experienced two different conditions (adaptive and static) (Charness *et al.*, 2012). Six participants started with the adaptive environment, while the remaining started with the static one; this counterbalancing procedure was implemented to prevent order and sequence effects (Corriero, 2017). In the static control condition, the lighting was fixed at maximum brightness and a cool colour temperature, and all audio tracks played continuously at a constant volume. These settings were selected to simulate the standard high-brightness, cool-tone lighting conditions commonly found in high street fitting rooms, as identified during user research.

B. Procedure

To simulate fitting room anxiety, participants were asked to solve complex mathematical tasks prior to each condition (Kirschbaum *et al.*, 1993). This brief stress induction raised their physiological baseline, allowing the study to test whether the adaptive system could actively support recovery from a pre-existing stressed state. Participants then entered the fitting room, where sensor connection and calibration required a further one to two minutes; because BLE connection time varied between sessions, this left an average of approximately three minutes of measured condition exposure per session, during which participants used a tablet to select and rate outfit combinations. This simulated natural fitting room behaviour while keeping cognitive demand low enough to avoid causing additional stress.

C. Measures

Subjective arousal was measured using the Self-Assessment Manikin (SAM), a non-verbal, visual scale that asks participants to rate their current state by selecting a manikin figure ranging from sleepy and relaxed to wide-eyed and very stimulated. SAM is a fast, intuitive, and validated tool that accurately captures emotional responses (Bradley and Lang, 1994). As it requires minimal completion time, it was ideal for capturing repeated measurements at three points: the true baseline before stress induction, immediately after the static condition, and immediately after the adaptive condition. The template of the questionnaire can be found under Appendix D.

To capture physiological data, participants wore the heart rate sensor throughout the entire procedure, enabling a direct comparison of physiological recovery between the adaptive and static environments. Unlike the subjective SAM measure, no separate resting physiological baseline was captured prior to the stress induction; instead, RR-interval calibration occurred at the start of each condition session via the system's built-in 20-beat calibration phase. Therefore, the physiological analysis reflects within-session RR-interval trajectory rather than comparison against a pre-stress resting baseline. This distinction is addressed further in the Limitations section.

D. Statistical Analysis

Given the modest sample size ($n=11$), a Bayesian rather than frequentist statistical approach was adopted. Traditional significance testing relies on large-sample assumptions and can be unreliable in finite samples, often overstating evidence against the null hypothesis (Rouder *et al.*, 2009). To overcome this, hypothesis testing was conducted using the Bayes Factor, which directly calculates the ratio of evidence between the alternative and null hypotheses (Rouder *et al.*, 2009). A Bayesian paired t-test compares two related measurements taken from the same participants (e.g., before and after a treatment) by testing whether the participant-level difference scores are more consistent with a null hypothesis of zero mean difference or an alternative hypothesis of nonzero difference (Gronau *et al.*, 2020; Rouder *et al.*, 2009). Rather than a p-value, the result is expressed as a Bayes Factor (BF_{10}), which measures the relative likelihood of the data under the alternative versus the null hypothesis and functions as an updating factor from prior odds to posterior odds (van Doorn *et al.*, 2021):

$$BF_{10} = \frac{P(\text{data} \mid H_1)}{P(\text{data} \mid H_0)}$$

Paired Bayesian t-tests were conducted using Pingouin's *bayesfactor_ttest()* function with its default settings. These settings use a widely-recommended assumption about how large the true effect size is likely to be before seeing the data (known as the JZS prior), which favours small-to-moderate effects while still allowing for larger ones (Rouder *et al.*, 2009, 2012). To determine the strength of the results, the Bayes Factors were evaluated using Jeffreys' (1961) classic evidence scale (Wetzels *et al.*, 2011).

E. Evaluation Participants

As in the user research phase, the evaluation sample focused on Gen Z individuals over 18 due to commercial reasons. Recruitment was conducted through the researcher's personal network and QMUL student contacts, introducing an element of convenience sampling which is acknowledged as a limitation. This likely resulted in a sample that is homogeneous in terms of geographic location and educational background, which limits the generalisability of the findings across the broader consumer market. As with the user research phase, ethical approval was obtained prior to data collection, and ethical documentation relevant to this stage can be found under Appendix E. Table II gives an overview of the 11 participants in this stage.

TABLE II. SYSTEM EVALUATION SAMPLE

Category	Details
Genders	5 Females, 6 Males
Age ranges	9 Gen Z's over 18 (born 1997-2007), 2 Millennials (born 1981-1996)
Occupations	6 Full-Time Employed, 5 students

VI. RESULTS

A. Subjective Measure: SAM Arousal Scale

Scores from SAM questionnaires supported *RQ1*. Out of the eleven participants, seven reported reduced arousal from their pre-session baseline following the adaptive condition, compared to only four following the static condition. The mean SAM change from baseline was -1.18 under the adaptive condition, indicating a reduction in perceived stress, compared to +0.55 under the static condition, indicating a slight increase in stress. At the individual level, nine of the eleven participants reported lower arousal in the adaptive environment compared to the static one. A Bayesian paired t-test on the change from baseline showed very strong evidence that the adaptive condition reduced subjective stress more effectively than the static condition, with $BF_{10}=53.882$, $t(10)=-4.81$.

B. Physiological measure: Heart rate (RR-interval)

Analysis of RR-intervals did not reveal physiological recovery in either environment. Mean RR-intervals decreased in both the adaptive (with mean $M=-16.2$ ms) and static ($M=-12.8$ ms) conditions, with a numerically larger decrease observed in the adaptive condition. This was confirmed by the Bayesian paired t-test comparing RR-interval change under the adaptive condition against the static condition ($BF_{10}=0.302$, $t(10)=-0.18$), indicating moderate evidence against a physiological effect at the group level.

C. System Proportionality (*RQ2*)

To quantify the proportionality of the controller's response, Pearson correlation coefficients were calculated between the PID reaction score and RR-interval deviation

from baseline for each participant. Correlations were consistently strong across the sample (mean $r = 0.98$, range 0.95-0.99, all $p < .001$), confirming that the controller produced a smooth, monotonic mapping from physiological deviation to environmental output rather than discrete or inconsistent transitions. As the reaction score is computed directly from the RR-interval deviation, this correlation confirms that the PID control function was implemented correctly, rather than representing an independently observed physiological relationship.

D. System latency

Software processing cycle times were logged across 2,625 heartbeat cycles across all eleven participants. The mean cycle time was 194.3 ms, well within the 400 ms threshold identified by Ng *et al.* (2020) as the upper limit for perceptible real-time response, with only 12 cycles (0.5%) exceeding this threshold. These findings confirm that the *asyncio* architecture successfully maintained near-real-time responsiveness throughout evaluation. Table III shows a summary of all results.

TABLE III. RESULTS SUMMARY

Measure	Adaptive	Static	BF_{10}	Result
Mean RR change (ms)	-16.2	-12.8	0.302	Moderate evidence against physiological effect
Mean SAM change	-1.18	+0.55	53.882	Very strong for $H1$
PID-RR correlation (r)	Mean $r=0.98$	-	-	Very strong for $H2$

VII. DISCUSSION

A. RQ1: Does the adaptive environment reduce perceived stress?

The subjective results partially support RQ1. As established in the previous section, nine of eleven participants reported lower SAM arousal scores following the adaptive condition compared to the static condition. This consistent pattern indicates that the bio-responsive environment produced a noticeable and positive effect on subjective stress

experience for most participants. As an illustrative case study rather than representative evidence, P04 showed the largest individual effect, reporting a four-point SAM reduction under adaptive compared to only one point under static, demonstrating that the system can successfully detect and respond to their elevated baseline arousal (SAM = 8 at pre-session baseline).

This dissociation between subjective and physiological stress measures is well documented in the broader stress literature (Campbell and Ehler, 2012), and the present results follow the same pattern. Participants experienced a decrease in RR-intervals (indicating higher physiological arousal) in both conditions, with a numerically larger decrease under adaptive ($M=-16.2$ ms) than static ($M=-12.8$ ms). While the adaptive environment produced a strong subjective benefit, the physiological data do not support a corresponding recovery effect, and the raw means if anything point in the opposite direction (though this direction flips if one outlier, P09, is excluded). However, given the moderate evidence against a group-level effect in either direction ($BF_{10}=0.302$), this numerical difference should not be over-interpreted as a reliable finding. The lack of group-level physiological recovery is primarily attributable to the constraints of the brief evaluation window (see Limitations), compounded by the small sample size ($n=11$), which severely limited statistical power.

B. RQ2: Does the PID controller produce a proportional physiological response?

Session log data directly supports RQ2. The PID controller produced reaction scores across its theoretical range of -100 to +100, demonstrating that the system was sufficiently sensitive to detect both acute stress spikes and gradual recovery within the evaluation window. Rather than a uniform shift toward calm, individual trajectories varied (Fig. 5 shows the smoothest reactions-score trajectories, selected for visual clarity): some participants (e.g. P01) showed a broadly upward trend, while others (e.g. P10) exhibited a pronounced late-session stress dip, indicating the controller remained responsive to real-time fluctuations throughout the session rather than settling into a fixed state. Although eight of the 11 participants achieved a positive mean PID score, this represents an overall trajectory of recovery rather than a linear progression in every session.

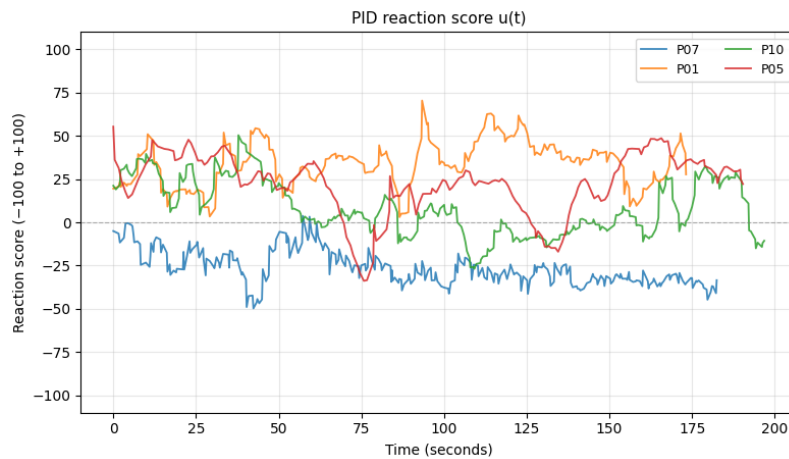


Fig. 5. PID reaction score trajectories for four participants.

The session logs confirm this proportional response: P01's reaction score oscillated throughout the session while remaining consistently on the calm side, whereas P07

remained on the stressed side for the entire session, showing the controller tracked real physiological deviation rather than uniformly driving every participant toward calm. No

flickering was observed or reported during any evaluation session. This stability is attributable to the 8-beat rolling average, which removes beat-to-beat variation before it reaches the controller, and the incremental light update, which bounds the change per heartbeat. This technical stability is critical from a commercial and user-experience perspective: in a highly evaluative space like a fitting room, an environment that visibly flickers or overreacts to sudden data spikes would introduce new sensory stressors, directly undermining the calming objective of the intervention. Together with the quantitative correlation analysis reported in Section VI-C, these observations confirm *H2* and support *RQ2*: the PID architecture succeeded not just as a continuous control mechanism, but as a physiologically appropriate foundation for retail stress reduction.

VIII. CONCLUSION

This project presented the first prototype applying real-time, heart rate-driven multimodal biofeedback to a retail fitting room context. By synthesizing consumer research with affective computing principles, the project successfully fulfilled all foundational objectives. An asynchronous Python architecture was engineered to connect a Polar H10 sensor to environmental lighting and audio, successfully managing two simultaneous BLE connections alongside wired audio output while operating within the sub-400 ms latency threshold required for real-time responsiveness. Furthermore, the within-subjects evaluation confirmed the core hypothesis: the adaptive environment produced a significantly greater reduction in subjective stress than a static control. The dissociation between strong subjective recovery and null physiological change is itself a substantive finding, consistent with the existing literature as discussed in section VII-A, and indicates the two measures should be treated as complementary rather than redundant signals in future affective retail systems.

A. Limitations

Several methodological and statistical limitations should be acknowledged. The brief and inconsistent exposure window may have limited the detection of physiological effects as reliable HRV recovery measurement requires consistent and extended observation periods (Roddick *et al.*, 2025). This constraint was compounded by variable BLE connection times, which meant the actual measured exposure window varied around approximately three minutes rather than consistently matching the originally intended five minutes, further limiting the detection of physiological effects. Furthermore, the study lacked a separate pre-stress baseline measurement. This omission occurred because the researcher's familiarity with the system during self-testing masked the need for an independent resting baseline. Consequently, physiological recovery could only be measured against the start of the session rather than the participant's true resting state. Statistically, the sample of 11 participants limits statistical power, and recruitment through the researcher's personal network and QMUL student contacts introduced convenience sampling bias, both of which limit the generalisability of findings.

In addition to methodological constraints, the study faced ecological and practical limitations. Participants did not physically try on garments, limiting the ecological validity of the evaluation. Additionally, the Polar H10 chest strap, while providing high-fidelity RR-interval data, presents practical barriers in a real retail fitting room context where users would

need to undress; consumer smartwatches may be a more practical alternative, although they commonly use photoplethysmography (PPG) sensors, which are less accurate than ECG devices, especially during movement (Georgiou *et al.*, 2018). Finally, the requirement for participants to remain stationary during sessions to avoid motion artefacts in RR-interval readings represents an ecological constraint that would not reflect natural fitting room behaviour, where movement during garment try-on is unavoidable.

IX. FUTURE WORK

Several clear directions for future research emerge from this study. Methodologically, future evaluations must address the constraints of the current experimental design. Specifically, exposure windows should be extended significantly beyond the approximately three-minute window achieved in this study to allow slower physiological HRV recovery to manifest. This extended timeframe should be coupled with a larger and more demographically diverse sample to achieve the statistical power necessary to detect group-level physiological effects. Moreover, the breathing modulation layer showed informally observed responsiveness during sessions and would benefit from formal evaluation in future work, e.g. using a respiration belt to quantify synchronisation accuracy.

Technically, the bio-responsive architecture can be advanced across both sensory modalities. The dynamic stem-mixing audio engine could be expanded into fully generative HRV-to-music synthesis, as demonstrated in domestic settings by Idrobo-Ávila *et al.* (2022). The current four-stem composition and mixing choices were made without formal training in music production or sound design, and future work would benefit from collaboration with an audio/music specialist to refine the stems' composition and how they layer. Furthermore, migrating the lighting output from a warm/cool white axis to full RGB control would enable the use of specific, calming colour palettes, allowing for a more personalized and immersive stress-reduction environment. Moreover, while the incremental update mechanism described at the end of section IV reliably prevented abrupt transitions, further tuning the Derivative gain could reduce reliance on this separate safeguard, though this would need to be balanced against maintaining responsiveness to genuine acute stress events. To improve system accuracy, machine learning could be used to customize PID parameters for individual users during calibration.

Finally, to bridge the gap between experimental prototyping and commercial application, future iterations must be deployed in an active retail fitting room. Testing the system while users engage in physical garment try-on will address the ecological validity limitations of this evaluation and confirm the commercial viability of affective AmI in fashion retail. However, this will require migrating the system from a laptop-based controller to embedded hardware.

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Participant Information Sheet

Title

MSc Project: **The Empathetic Fitting Room: A Physiologically Adaptive Retail Solution**

Version number and date:

Version 1.0: 14/06/2026

Researcher's name

Supervisor: Dr Fabrizio Smeraldi
Researcher: Agnese Enrica Picchio

Queen Mary Research Ethics Committee reference number:

DSEEC26.048

Invitation paragraph

You are being invited to participate in a research study.

Before you decide whether you wish to participate in this study, it is important for you to understand why the research is being done and what it will involve. Please take time to read the following information carefully and discuss it with others if you wish. Ask me any questions if there is anything that is not clear or if you would like more information. Taking part in this research study is entirely voluntary - refusal to participate requires no reason and will not affect you or your rights.

What is the purpose of the study and what would taking part involve?

As part of an MSc project in the School of Electronic Engineering and Computer Science at QMUL, this study investigates how "smart" environments can reduce stress during the retail shopping experience. Specifically, I am exploring a bio-responsive fitting room that automatically adjusts lighting and music based on a user's real-time heart rate.

You are invited to a shopping experience interview. This involves a 20–30 minute semi-structured interview (conducted in person on campus or online via Teams) discussing your views, experiences, and frustrations with physical retail fitting rooms.

Why am I being invited?

You are being invited to participate because you are an adult (aged 18 or over) who has experience shopping for clothes in physical retail stores. Please note we reserve the right to remove anyone from the study who is not a genuine participant. This includes, but is not limited to, misrepresenting eligibility. The decision is at the researchers' discretion.

Do I have to take part?

This participant information sheet has been written to help you decide if you would like to take part. It is entirely up to you. All questions and activities as part of this study are voluntary. You can choose not to answer any individual question or stop the session at any time, without question and without explanation.

What are the possible benefits of taking part?

There may be no direct personal benefit from taking part in this study. However, you may find it interesting to reflect on your shopping experiences. Your contribution will directly aid in the design of a system aimed at improving mental well-being in retail environments.

What are the possible disadvantages and risks of taking part?

The risks associated with taking part in this study are minimal. During Phase 1, you may find that recalling negative shopping experiences causes mild discomfort. You may stop your participation at any time without giving a reason.

Expenses and payments

We do not expect you to incur any costs while participating in this study. There will be no reimbursement for your time.

What information about me will you be collecting?

We will collect qualitative information about your views and experiences. With your explicit consent, we will take an audio recording of the interview to allow for accurate transcription. We may ask optional background questions, but this information will not identify you personally. Participants will not be identifiable in published or shared reports.

How will my data be stored and who will have access to it?

Your data will be stored securely on a password-protected local drive and QMUL's OneDrive file storage system. For Phase 1, the original audio recordings will be permanently destroyed the exact moment they are typed up into text. Only the researcher and supervisor listed at the start of this document will have access to the raw data.

When and how will my data be destroyed?

All data will be retained securely for the duration of the project. In accordance with the QMUL Records Retention Schedule and data protection principles, data will not be kept for longer than is necessary. As stated above, audio recordings are destroyed immediately after transcription. All remaining anonymised data, such as the written text transcripts, will be permanently and securely destroyed following the final submission and Viva of the MSc dissertation (September 2026).

How will my data be used and shared?

Data will be analysed to inform the design of the prototype. The results of this analysis will be included in a final MSc dissertation graded by staff members at QMUL. As previously stated, you will not be identifiable in the final dissertation or any resulting publications.

Under what legal basis are you collecting this information?

- Queen Mary University of London **processes personal data** for research purposes in accordance with the lawful basis of 'public task'.
- Please read [Queen Mary's privacy notice for research participants](#) containing **important information about your personal data and your rights** in this respect.
- **If you have any questions** relating to data protection, please contact Queen Mary's Data Protection Officer, Queens' Building, Mile End Road, London, E1 4NS or data-protection@qmul.ac.uk or 020 7882 7596.

What will happen if I want to withdraw from this study?

You can withdraw from the study at any time during the session without providing a reason, and this will not have any effect on you or your rights. After the interview, you may request to withdraw your data up to 7 days after your session by providing the researcher with your unique participant code. After this timeframe, the audio recording will be transcribed and permanently destroyed, meaning your data will become fully anonymized and can no longer be identified for withdrawal.

What should I do if I have concerns about this study?

- **If you have any concerns** about the manner in which the study was conducted, in the first instance, please **contact the researcher(s)** responsible for the study, Dr Fabrizio Smeraldi (f.smeraldi@qmul.ac.uk).

- **If you have a complaint** which you feel you cannot discuss with the researchers then you should **contact the Research Ethics Facilitators** by e-mail: research-ethics@qmul.ac.uk.
- When contacting the Research Ethics Facilitators, **please provide details** of the study title, description the study and QM REC reference number (where possible), the researcher(s) involved, and details of the complaint you wish to make.

Who can I contact if I have any questions about this study?

Agnese Enrica Picchio

ec25976@qmul.ac.uk

Consent Form

Title of Research Study:

MSc Project: **The Empathetic Fitting Room: A Physiologically Adaptive Retail Solution**

Principal Investigator:

Supervisor: Dr Fabrizio Smeraldi

Researchers: Agnese Enrica Picchio

Queen Mary Research Ethics Committee reference number:

DSEECS26.048

Thank you for your interest in this research.

Should you wish to participate in the study, **please consider the following statements.**

Before signing the consent form, you should **initial all or any of the statements that you agree with.**

Your signature **confirms that you are willing to participate in this research**, however you are reminded that **you are free to withdraw your participation** at any time.

Statement	Please initial box
1. I confirm that I have read the Participant Information Sheet dated 14/06/2026 version 1.0 for the above study; or it has been read to me. I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily.	
2. I understand that my participation is voluntary and that I am free to stop taking part in the study at any time during participation without giving any reason and without my rights being affected.	
3. I understand that my data will be accessed by the research team.	
4. I understand that my data will be securely stored on QMUL OneDrive and an encrypted local drive in accordance with the data protection guidelines of the Queen Mary University of London until September 2026 in fully anonymised form.	
5. I understand that I can withdraw my data from the study up to 7 days after my interview by providing the researcher with my unique participant code. After this timeframe, my audio recording will be transcribed and destroyed, and my data will become fully anonymized, meaning I will no longer be able to withdraw it.	
6. I agree to the interview being audio recorded for transcription purposes, and understand the audio file will be destroyed immediately after transcription.	

7. I understand that the researcher will not identify me in any publications and other study outputs using personal information obtained from this study.	
8. I agree to take part in the above study.	

Participants should read [Queen Mary's privacy notice for research participants](#) which contains **important information about your personal data and your rights** in this respect.

If you have any **questions relating to data protection**, please contact Data Protection Officer, Queens' Building, Mile End Road, London, E1 4NS or data-protection@qmul.ac.uk or 020 7882 7596.

_____	_____	_____
Participant name	Date	Signature

_____	_____	_____
Name of person taking consent	Date	Signature

I, Agnese Enrica Picchio, confirm that I have carefully explained the nature, demands and any foreseeable risks (where applicable) of the proposed research to the participant and provided a copy of this form.

Supervisor

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Student Investigator

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Appendix B – Interview guide

Research question	Interview phase	Specific interview question
RQ1: Does a bio-responsive environment adapting lighting and audio based on real-time HRV reduce perceived user stress compared to a static environment? <i>(Note: Primarily evaluated via the SAM scale during Phase 2, but these questions establish the baseline and perceived value).</i>	Warm-up & Fitting Room Experience	• How do you feel about using fitting rooms? Walk me through what the experience is typically like for you. • Tell me about the worst experience you've recently had in a fitting room. What exactly made it bad? • Tell me about the best or most premium fitting room experience you've had. Why did it stand out?
	Concept response	• If the environment around you automatically adjusted when you were feeling stressed (e.g. music getting slower, lighting getting warmer) - do you think that would actually help, or would you find it intrusive? • Fitting rooms are rarely completely isolated or soundproof. If the room's lights or music actively changed based on your state, would you worry about people outside knowing how you feel, or would you actually find it reassuring because it signals to others that the room is occupied? • Imagine a store offered this empathetic fitting room experience, but to use it, you had to put on a heart-rate monitor strap around your chest before trying on clothes. What is your immediate reaction to that? • How would you feel about a retail store collecting your real-time heart rate data, assuming it wasn't tied to your name? Where do you draw the line on privacy?
	Environmental Factors	• How does fitting room lighting affect how you feel about yourself and what you're trying on? If you could customize it, what specific type of lighting (e.g., warm, cool, dim, bright) would make you feel the most comfortable? • Have you ever felt overwhelmed or overstimulated by the atmosphere in a shop, e.g. music too loud, lighting too harsh? What was that like? • Does background music in shops affect your mood or behaviour? In what way? If a fitting room were trying to calm you down, what specific sounds or tempos would work best for you? • If you had a magic wand to change the sensory environment (specifically the

		lighting and audio) of a standard fitting room, what is the exact combination you would create?
	Concept response	How would you feel about music that was being created in real time just for you from your heartbeat, rather than songs you already know?
RQ2: Does a continuous adaptive control architecture produce a proportional environmental response that reflects the user's physiological stress trajectory throughout the session?	System Evaluation & Qualitative Support	- Quantitative System Logs (Primary): Evaluated via PID controller reaction scores and HRV trajectory data across the 3-minute sessions. - Concept Response (Qualitative Support): How would you feel about music that was being created in real time just for you from your heartbeat, rather than songs you already know? (Used to analyze user feedback on the perceived "smoothness" and real-time continuity of the system).

Appendix C – Thematic analysis

Theme	Codes	Definition of theme
The Fitting Room as a Chore	• Queuing and waiting times • Feeling rushed or pressured • Lack of space/missing hooks • Poor hygiene and messiness • Flimsy curtains/lack of privacy	Fitting rooms are widely perceived as stressful, practical tasks rather than enjoyable experiences. This stress is heavily exacerbated by physical inconveniences, crowded spaces, and a lack of secure privacy.
Sensory Overload in Retail Spaces	• Loud, fast-paced music • Crowds and noisy people • Overpowering artificial smells • Cluttered visual environment	Harsh and aggressive environmental stimuli negatively impact shoppers' moods, leading to overstimulation, anxiety, and a strong desire to abandon the shopping experience and leave the store entirely.
The Psychological Impact of Lighting	• Unflattering top-down angles • Distorted body image • Preference for warm/natural tones • Influence on purchase decisions	Lighting acts as a critical mechanism for self-perception. Poor lighting triggers severe body image insecurities and emotional distress, directly resulting in abandoned purchases.
The Wearable Tech Barrier	• Impracticality of chest straps • Preference for wrist wearables • Value-Exchange Acceptance	Multiple users noted that wearing a chest strap while changing clothes is physically impractical compared to a wristband. However, users stated they would tolerate the device if they understood its purpose, or received a highly personalized, premium sensory payoff.
Data Privacy and Transparency	• Anonymity Prerequisite • Demand for Consent/Transparency • Apathy towards De-identified Data • Distrust of Unexplained Sensors	Users expressed initial curiosity or confusion about heart-rate tracking, but uniformly agreed they had zero concerns regarding data privacy as long as the biometric data was strictly anonymous.

Appendix D – SAM questionnaire

Participant code	Session date	Condition order (S→A / A→S)

Instructions: For each timepoint, tick the box that best describes reflects your current state. Please respond based on your immediate feeling, not how you think you should feel. Each scale goes from 1 (very calm) to 9 (very tense).

Baseline: Before entering the room

AROUSAL

1 = Completely calm and relaxed 9 = Very stimulated and tense

								
1	2	3	4	5	6	7	8	9

After Condition A

AROUSAL

1 = Completely calm and relaxed 9 = Very stimulated and tense

								
1	2	3	4	5	6	7	8	9

After Condition B

AROUSAL

1 = Completely calm and relaxed 9 = Very stimulated and tense

								
1	2	3	4	5	6	7	8	9

Participant Information Sheet

Title

MSc Project: **The Empathetic Fitting Room: A Physiologically Adaptive Retail Solution**

Version number and date:

Version 1.0: 14/06/2026

Researcher's name

Supervisor: Dr Fabrizio Smeraldi
Researcher: Agnese Enrica Picchio

Queen Mary Research Ethics Committee reference number:

DSEEC26.048

Invitation paragraph

You are being invited to participate in a research study.

Before you decide whether you wish to participate in this study, it is important for you to understand why the research is being done and what it will involve. Please take time to read the following information carefully and discuss it with others if you wish. Ask me any questions if there is anything that is not clear or if you would like more information. Taking part in this research study is entirely voluntary - refusal to participate requires no reason and will not affect you or your rights.

What is the purpose of the study and what would taking part involve?

As part of an MSc project in the School of Electronic Engineering and Computer Science at QMUL, this study investigates how "smart" environments can reduce stress during the retail shopping experience. Specifically, I am exploring a bio-responsive fitting room that automatically adjusts lighting and music based on a user's real-time heart rate.

You are invited to take part in a Prototype Evaluation. This involves a 30-minute session at a designated space on the QMUL campus. You will spend a short amount of time inside a physical mockup of the "smart" fitting room to evaluate how the changing lights and audio affect your mood. To allow the room to respond to your physiological signals, you will be asked to wear a standard consumer chest-strap heart rate monitor (Polar H10) for the duration of the session. Please note: you will NOT be asked to try on any clothes, nor will you be asked to remove any of your own clothing, veils, or garments during this study.

Why am I being invited?

You are being invited to participate because you are an adult (aged 18 or over) who has experience shopping for clothes in physical retail stores. If you are participating in Phase 2, you must also be comfortable wearing a chest-based heart rate monitor.

You **should not take** part in this study if you have any diagnosed cardiac conditions that might affect heart rate variability readings, or if you are unable to provide informed consent.

Please note we reserve the right to remove anyone from the study who is not a genuine participant. This includes, but is not limited to, misrepresenting eligibility. The decision is at the researchers' discretion.

Do I have to take part?

This participant information sheet has been written to help you decide if you would like to take part. It is entirely up to you. All questions and activities as part of this study are voluntary. You can choose not to answer any individual question or stop the session at any time, without question and without explanation.

What are the possible benefits of taking part?

There may be no direct personal benefit from taking part in this study. However, you may find it interesting to interact with a novel piece of Affective Computing technology. Your contribution will directly aid in the evaluation of a system aimed at improving mental well-being in retail environments.

What are the possible disadvantages and risks of taking part?

The risks associated with taking part in this study are minimal.

During the evaluation, you will be required to wear a fitness chest strap, which some may find slightly unfamiliar, though it is a standard, non-invasive consumer device.

You may stop your participation at any time without giving a reason.

Please note: The Polar H10 chest-strap heart rate monitor is being used strictly for research purposes. It is not being used for medical diagnosis, and you will not receive any health feedback or medical assessments based on the data collected.

Expenses and payments

We do not expect you to incur any costs while participating in this study. There will be no reimbursement for your time.

What information about me will you be collecting?

We will collect your raw Electrocardiogram (ECG) and beat-to-beat interval data via the Polar H10 sensor. This data will be used to calculate your Heart Rate Variability (HRV), filtering out any signal artifacts or ectopic heartbeats, alongside your responses to a short, printed questionnaire measuring your perceived stress levels (the SAM scale). We may ask optional background questions, but this information will not identify you personally. Participants will not be identifiable in published or shared reports.

How will my data be stored and who will have access to it?

Your data will be stored securely on a password-protected local drive and QMUL's OneDrive file storage system. Only the researcher and supervisor listed at the start of this document will have access to the raw data.

When and how will my data be destroyed?

All data will be retained securely for the duration of the project. In accordance with the QMUL Records Retention Schedule and data protection principles, data will not be kept for longer than is necessary. All anonymised data, such as your physiological HRV logs and questionnaire responses, will be permanently and securely destroyed following the final submission and VIVA of the MSc dissertation (September 2026).

How will my data be used and shared?

Data will be analysed to evaluate the effectiveness of the prototype. The results of this analysis will be included in a final MSc dissertation graded by staff members at QMUL. As previously stated, you will not be identifiable in the final dissertation or any resulting publications.

Under what legal basis are you collecting this information?

- Queen Mary University of London **processes personal data** for research purposes in accordance with the lawful basis of 'public task'.
- Please read [Queen Mary's privacy notice for research participants](#) containing **important information about your personal data and your rights** in this respect.
- **If you have any questions** relating to data protection, please contact Queen Mary's Data Protection Officer, Queens' Building, Mile End Road, London, E1 4NS or data-protection@qmul.ac.uk or 020 7882 7596.

What will happen if I want to withdraw from this study?

You can withdraw from the study at any time during the session without providing a reason, and this will not have any effect on you or your rights. After the evaluation, you may request to withdraw your data up

to 7 days after your session by providing the researcher with your unique participant code. After this timeframe, your physiological data and questionnaire responses will be fully anonymized and can no longer be identified for withdrawal.

What should I do if I have concerns about this study?

- **If you have any concerns** about the manner in which the study was conducted, in the first instance, please **contact the researcher(s)** responsible for the study, Dr Fabrizio Smeraldi (f.smeraldi@qmul.ac.uk).
- **If you have a complaint** which you feel you cannot discuss with the researchers then you should **contact the Research Ethics Facilitators** by e-mail: research-ethics@qmul.ac.uk.
- When contacting the Research Ethics Facilitators, **please provide details** of the study title, description the study and QM REC reference number (where possible), the researcher(s) involved, and details of the complaint you wish to make.

Who can I contact if I have any questions about this study?

Agnese Enrica Picchio

ec25976@qmul.ac.uk

Consent Form

Title of Research Study:

MSc Project: **The Empathetic Fitting Room: A Physiologically Adaptive Retail Solution**

Principal Investigator:

Supervisor: Dr Fabrizio Smeraldi

Researchers: Agnese Enrica Picchio

Queen Mary Research Ethics Committee reference number:

DSEECS26.048

Thank you for your interest in this research.

Should you wish to participate in the study, **please consider the following statements.**

Before signing the consent form, you should **initial all or any of the statements that you agree with.**

Your signature **confirms that you are willing to participate in this research**, however you are reminded that **you are free to withdraw your participation** at any time.

Statement	Please initial box
1. I confirm that I have read the Participant Information Sheet dated 14/06/2026 version 1.0 for the above study; or it has been read to me. I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily.	
2. I understand that my participation is voluntary and that I am free to stop taking part in the study at any time during participation without giving any reason and without my rights being affected.	
3. I understand that my data will be accessed by the research team.	
4. I understand that my data will be securely stored on QMUL OneDrive in accordance with the data protection guidelines of the Queen Mary University of London until September 2026 in fully anonymised form.	
5. I understand that I can withdraw my data from the study up to 7 days after my evaluation session by providing the researcher with my unique participant code. After this timeframe, my physiological data and questionnaire responses will be fully anonymized, meaning I will no longer be able to identify and withdraw it.	
6. I agree to wear the Polar H10 chest-strap heart rate monitor for the duration of the evaluation session to allow the collection of Electrocardiogram (ECG) and Heart Rate Variability (HRV) data.	

7. I understand that the researcher will not identify me in any publications and other study outputs using personal information obtained from this study.	
8. I agree to take part in the above study.	

Participants should read [Queen Mary's privacy notice for research participants](#) which contains **important information about your personal data and your rights** in this respect.

If you have any **questions relating to data protection**, please contact Data Protection Officer, Queens' Building, Mile End Road, London, E1 4NS or data-protection@qmul.ac.uk or 020 7882 7596.

_____	_____	_____
Participant name	Date	Signature

_____	_____	_____
Name of person taking consent	Date	Signature

I, Agnese Enrica Picchio, confirm that I have carefully explained the nature, demands and any foreseeable risks (where applicable) of the proposed research to the participant and provided a copy of this form.

Supervisor

Dr Fabrizio Smeraldi

f.smeraldi@qmul.ac.uk

Student Investigator

Agnese Enrica Picchio

ec25976@qmul.ac.uk

ECS750P/ECS753P/ECS754P– MSc Project Module
Generative AI Tools – Student Accountability Statement

The completed and signed statement must be included in the appendix of the final dissertation paper.

Student Name: **Agnese Enrica Picchio**

Project Title: The Empathetic Fitting Room: **A Physiologically Adaptive Retail Solution**

Supervisor Name: **Dr Fabrizio Smeraldi**

1. Have you used Generative AI tools for your project?

YES

NO

If yes, please complete sections 2 and 3 and sign the declaration below.

If no, please sign the declaration below.

2. Please provide details of all the AI tools that you used in your project work.

AI Tool	Version	Publisher	Access URL
Claude	Sonnet 5	Anthropic	https://claude.ai
Gemini	3.1 Pro	Google	https://gemini.google.com/

3. If you have used Artificial Intelligence (AI) tools in any part of your project please include in table format how, where, and why these tools were used. This should include references to the relevant parts of your project work — for example:

- Specific chapters or sections of the report (e.g. literature review, methodology, requirements)
- Software development (e.g. AI-assisted code generation, debugging, documentation)
- Interview or survey materials (e.g. AI-generated or refined questions/prompts)
- Datasets (e.g. AI-assisted data cleaning, labelling, or synthesis)
- The showcase video (e.g. AI-assisted scripting, narration, or editing support)

Be specific and transparent about your use of AI tools, including both direct outputs (e.g. text, code, images) and indirect uses (e.g. idea generation, editing

assistance). The goal is to ensure academic integrity and provide a clear record of how AI contributed to your work.

Project Section	Description	Justification
System Design (Live Monitor Interface)	An initial version of the real-time monitoring interface was generated using Claude, based on prompts specifying the required data fields and the constraint of running as a separate process from the <code>asyncio</code> pipeline. This AI-generated version was then used as a reference during development of the final interface used in evaluation sessions.	AI was used to prototype the initial graphical interface. Because this is an auxiliary component, using AI freed up valuable time to focus on the main system and evaluation. This file is marked with a header comment.
Software Debugging	Claude assisted in diagnosing and resolving specific technical issues, including a BLE timeout in an <code>asyncio</code> task and a missing class attribute in the PID controller.	This accelerated troubleshooting in asynchronous code. All AI-suggested fixes were carefully reviewed, tested, and fully understood by the researcher prior to implementation.
Writing and editing	Gemini was utilized to refine prose, improve sentence clarity, ensure adherence to the 8-page limit, and verify terminological consistency across the dissertation and reflective essay.	AI served as an editing support for clarity and concision. All original analysis, arguments, findings, and interpretations remain entirely the researcher's own.

System Design - Control Strategy Comparison (Fig. 4)	Claude was consulted for guidance on matplotlib techniques (e.g. subplot creation, shared axis formatting, figure export) for the implementation of <i>plot_comparison_grap</i> <i>hs.ipynb</i> , but all project-specific content, including the underlying data, colour thresholds, axis labels, and analytical annotations, reflects the researcher's own understanding.	AI assistance accelerated general plotting/formatting technique for a figure whose analytical content is the researcher's own. All plotting logic was reviewed, adapted, and is fully understood by the researcher. This file is marked with a header comment disclosure.
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3. Generative AI Student Declaration

- I have declared all uses of AI tools (e.g., coding assistants, text generators, data analysis tools) in this project.
- I have reviewed and verified all AI-assisted content to ensure accuracy and suitability.
- I take full responsibility for the originality and integrity of my project submission.
- I have referenced all sources, including those suggested or produced with AI support.
- I have not used AI in ways that undermine academic honesty (e.g., plagiarism, unverified code submission).
- My use of AI was supplementary and did not replace my own independent learning and problem-solving.

Signed: Agnese Enrica Picchio Date: 16/08/2026

Appendix G – Project Proposal

**School of Electronic Engineering and Computer Science
Queen Mary University of London**

**MSc PROJECT DEFINITION
2025-26**

This project definition must be undertaken in consultation with your supervisor. The feasibility of the project should have been assessed and the project aims should be clearly defined.

Submission of this document implies that you have discussed the specification with your supervisor.

Project Title: The Empathetic Fitting Room: A Physiologically Adaptive Retail Solution

Supervisor: Dr Fabrizio Smeraldi

Student name: Agnese Enrica Picchio

Student e-mail: ec25976@qmul.ac.uk

PROJECT AIMS:

The commercial rationale of this project comes from an industry need to improve the physical shopping experience, particularly for younger demographics like Gen Z. Recent data highlights traditional fitting rooms as a major friction point: 39% of shoppers experience "fitting room anxiety," and 36% leave stores without purchasing due to negative in-store experiences [1]. Since retail purchases are mainly driven by emotion [2], transforming stores into "sensory destinations" [3] with carefully designed lighting and sound can dramatically improve customer behavior.

This project aims to create an innovative solution to this problem by creating a bio-responsive fitting room. The main technical challenge is building a highly reliable, real-time system that connects a user's physical stress levels to digital hardware. To achieve this without system delays, the project will utilize an asynchronous software architecture: it will continuously process live heart rate data via Bluetooth (using *bleakheart*) and immediately translate the physiological stress levels into synchronized adjustments to smart lighting and audio, creating a calming, personalized environment.

PROJECT OBJECTIVES:

- Objective 1: Linking a Polar wearable sensor to the main software using the *bleakheart* driver. The project will use this tool to track the precise timing between heartbeats (RR intervals) to calculate heart rate variability, serving as a measurement of the user's stress levels.
- Objective 2: Setting up a smooth, real-time data flow. This will allow the heart rate sensor to instantly pass its readings to the smart lights and audio system, so that the system can process live physical responses and change the room's environment simultaneously, without the software ever freezing or lagging.
- Objective 3: Designing the core mapping algorithm that translates physiological data into specific environmental outputs (light and sound variations). This algorithm

will first be developed and validated using a mockup with offline datasets. Once stabilized, the system will transition to live data acquisition.

- Objective 4: Designing the logic that translates the user's physical state into physical room changes. This involves choosing the best communication method for the smart lights and generating sounds or music that adapt to the user's heartbeat.
- Objective 5: Defining the guidelines for user testing. This includes building a physical mockup and obtaining University Ethical Approval to test the prototype with real people (if needed).

State how your project will be aligned with the learning outcomes of your programme of study.

This project meets the MSc program's goals by integrating computer science with cognitive psychology to solve a real commercial problem.

- Academic Content (A1, A2, A3): The project applies advanced computer science theories by using Python programming to build a responsive, multi-tasking system. Testing will also be used to evaluate how quickly and accurately lights and audio react to the user's live biological signals without noticeable delay.
- Disciplinary Skills (B1, B2, B3): The core challenge demonstrates the ability to design, build, and test a complete system that bridges the gap between software and physical hardware. It requires managing a flow of live sensor data and evaluating different wireless networking options to determine the most reliable way for the devices to communicate.

Furthermore, this prototype serves as a practical application of modules such as Computer Programming (managing multiple simultaneous tasks), Computer Architecture & Networks (connecting smart devices), Data Analytics (filtering and processing live biological data), and Interactive System Design.

METHODOLOGY:

1. Instead of conducting user interviews, the project will rely on extensive literature review to inform the initial design. This phase will give an overview of the general architecture of the system and focus on experience design, planning how a user will feel and interact within the bio-responsive fitting room.
2. Configuring the Polar sensor to ensure a Bluetooth connection using the *bleakheart* tool. This ensures the system can accurately and consistently read the user's live physical signals.
3. Developing the core software that allows the system to multitask smoothly. This framework will manage the incoming heartbeat data while instantly sending out commands to the lights and audio, ensuring the environment reacts in real-time without the software lagging or freezing.
4. Creating mathematical filters to clean up the raw, noisy sensor data, establishing a reliable baseline for the user's heart rate. Next, integrating the pyo audio software to generate or change ambient sounds and music so they naturally synchronize with the user's calming heartbeat.
5. Building the physical fitting room mockup. Connecting the main software to the smart LED lighting system, testing different communication methods (such as direct Bluetooth

or smart home hubs like Zigbee) to find the fastest, most reliable way to adjust the room's lighting based on the user's stress levels.

6. Conducting testing with participants in the mockup environment. This primary research has two main goals: to measure system efficiency (using the bleakheart tool's precise timing to ensure there is zero noticeable delay between a physical heartbeat and the room's reaction) and to evaluate the overall experience design, validating whether the system successfully creates a calming atmosphere.

7. Develop the core data-mapping algorithm. Initially, use offline heart rate datasets to simulate and validate how the logic triggers lighting and sound variations. This ensures the mathematical filters accurately clean data before testing with live subjects.

PROJECT MILESTONES

1. Milestone 1: Script to connect the wearable sensor via Bluetooth and extract live heart rate data. Submitting all required documentation for University Ethical Approval.
2. Milestone 2: Implementing the core software infrastructure that allows data to flow without lagging, and the algorithms needed to clean up noisy heart rate readings.
3. Milestone 3: Demonstration of the mapping algorithm, showcasing the ability to map pre-selected music and lighting parameters to simulated offline biometric data.
4. Milestone 4: Successful transition to live acquisition via bleakheart and integration of the physical IoT LED lights and pyo audio generation.
5. Milestone 5: Setting up the complete interactive prototype within the physical mockup of the fitting room, conducting primary user testing, analyzing the collected data, and drafting the final MSc dissertation.

REQUIRED KNOWLEDGE/ SKILLS/TOOLS/RESOURCES:

- Advanced Python programming
- Asynchronous programming paradigms (asyncio)
- Signal processing basics
- Audio synthesis logic
- IoT networking (evaluating Bluetooth vs. Wi-Fi/Zigbee APIs)
- Python 3
- Bleak library (for both Polar sensor input and potential HueBLE output)
- Bleakheart driver
- Pyo library (for asynchronous audio processing)
- Pandas (for data structuring)
- Polar sensor
- Bluetooth-enabled development machine
- Controllable Smart Lighting system capable of BLE or HTTP/Zigbee communication (e.g. Philips Hue)

TIMEPLAN



REFERENCES

- [1] "Fitting room frustration: New research reveals low confidence among fashion shoppers", *Zalando*, Oct. 8, 2024. Accessed: Mar. 26, 2026. [Online]. Available: <https://corporate.zalando.com/en/fashion/fitting-room-frustration-new-research-reveals-low-confidence-among-fashion-shoppers>
- [2] Mood Media, "Emotions drive sales: The power of behavioral economics", *Retail Dive*, Mar. 13, 2023. Accessed: Mar. 26, 2026. [Online]. Available: <https://www.retaildive.com/spons/emotions-drive-sales-the-power-of-behavioral-economics/643967/>
- [3] T., Sender Ceron, "Clothing Retailing – UK – 2025", Mintel, Nov. 28, 2025. Accessed: Mar. 26, 2026. [Online] Available: <https://clients.mintel.com/content/report/clothing-retailing-uk-2025?fromSearch=%3FcontentType%3DReport%26filters.category%3D122%26freetext%3Dretail%2520tech%26resultPosition%3D8>